Real Analysis

Motivation in Research

firstname lastname

Mumbai University

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Axioms of the Real Numbers

1. **Field axioms:** $\mathbb{R}$ has addition and multiplication that are commutative, associative and distributive, with identities 0 and 1 and inverses.
2. **Order axioms:** For any $x, y \in \mathbb{R}$ exactly one of $x < y$, $x = y$, $x > y$ holds, and the order respects addition and multiplication by positive numbers.
3. **Completeness axiom:** Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound (supremum).

References

 C. C. Pugh, *Real Mathematical Analysis*, Springer.

Thank You

I thank the Department of Mathematics for this opportunity.